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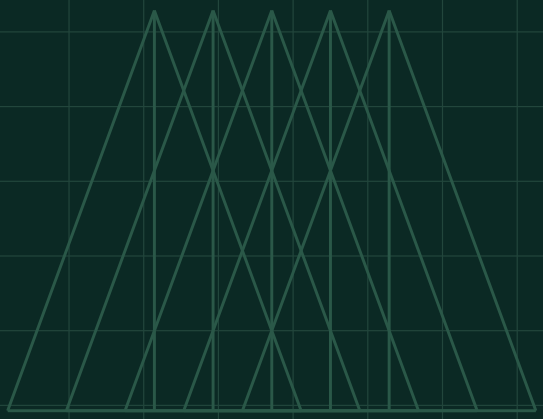
STRATEGIC RESEARCH / BRITISH AGRI-FOOD

BRITAIN'S AGRI-FOOD TRADE IMBALANCE

Investment priorities for a new generation of British growing infrastructure



An Agrispot investment briefing on where greenhouses, hydroponics and vertical farming can realistically reduce import dependence - and where they cannot.



£67.8bn

FOOD, FEED & DRINK IMPORTS

UK, 2025 current prices

£25.7bn

FOOD, FEED & DRINK EXPORTS

UK, 2025 current prices

£42.1bn

NET TRADE GAP

Arithmetic difference

Prepared for Agrispot | 13 October 2026

DECISION BRIEF • UK

Official trade statistics: Defra / HMRC. All recommendations are subject to project-level feasibility.

INVESTMENT THESIS / 01

From import exposure to investable production

Britain's food trade deficit is large, but only part of it is addressable through controlled-environment agriculture (CEA). Agrispot should focus on crops that can be grown competitively in the UK, with imported tropical commodities approached through supply-chain resilience and domestic processing.

FOOD, FEED & DRINK IMPORTS	FRESH FRUIT IMPORTS	DOMESTIC VEGETABLE SHARE
£67.8bn	**£4.99bn**	**56%**
2025, current prices [1]	2025 value; up 9.1% year on year [2]	Of total UK supply in 2025 [2]

What Agrispot should do first

RANK	TARGET	BEST-FIT SYSTEM	INVESTMENT LOGIC
1	Tomatoes, cucumbers, peppers	Modern glasshouses; substrate hydroponics	Established demand, technical maturity and a clear import-replacement opportunity in selected seasons.
2	Strawberries and raspberries	Polytunnels, glasshouses, substrate systems	High-value produce; protection can extend seasons and improve consistency.
3	Salad leaves and herbs	Hydroponic greenhouse; selective vertical farms	Short cycles and proximity to retail / food-service customers can support premium value.
4	Mushrooms	Controlled growing rooms	Compact, indoor production without the lighting burden of leafy-green vertical farms.
5	Coffee and cocoa	Diversified sourcing, UK processing; small research trials only	Large import category, but tropical growing economics make mass UK greenhouse production implausible today.

The central investment rule

Do not select a crop because its import bill is large. Select it because its UK production economics, seasonal advantage, buyer demand and supply-chain benefits justify the capital cost. CEA is a production tool, not a universal import-substitution technology.

Decision in one sentence

Build an Agrispot pipeline around protected vegetables, soft fruit and selected hydroponic crops; use transparent energy and offtake tests to screen investments; address tropical commodity exposure through supply diversification and higher-value British processing.

THE MARKET / 02

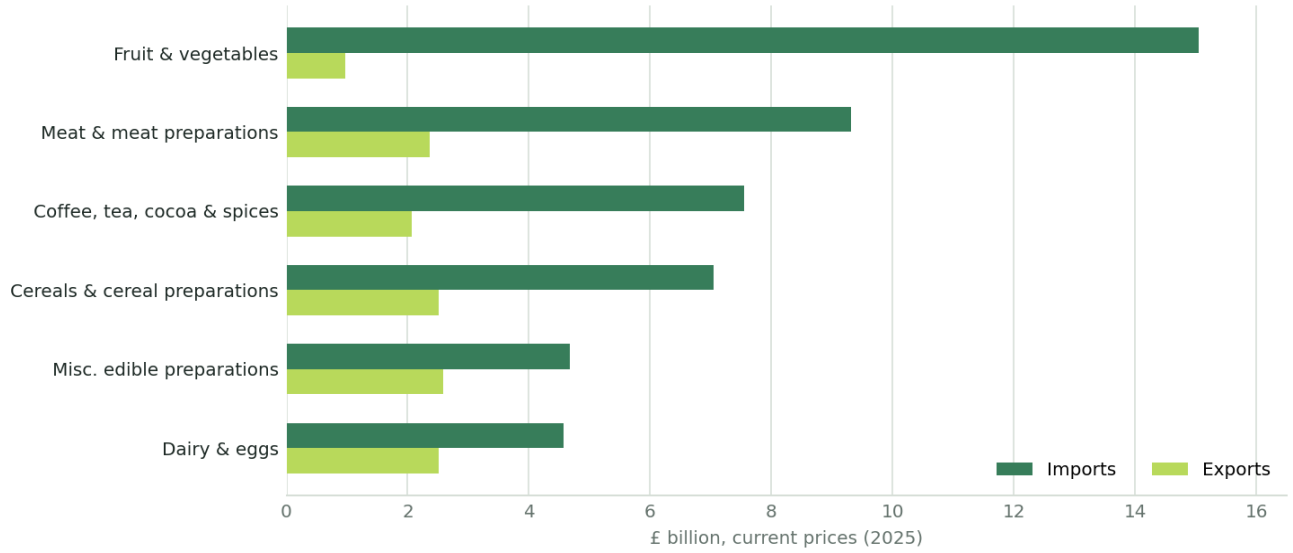
Where the import imbalance sits

The 2025 final trade release records £67.8bn of UK food, feed and drink imports and £25.7bn of exports: an arithmetic net gap of £42.1bn. Imports rose 6% in current prices and 5% in real terms compared with 2024. The figures cover a wide range of food products, not only farm commodities.

Selected commodity groups, 2025

COMMODITY GROUP	IMPORTS	EXPORTS	NET IMPORT GAP*
Fruit and vegetables	**£15.049bn**	£0.968bn	**£14.081bn**
Meat and meat preparations	**£9.314bn**	£2.363bn	**£6.951bn**
Coffee, tea, cocoa, spices, etc.	**£7.552bn**	£2.071bn	**£5.481bn**
Cereals and cereal preparations	**£7.046bn**	£2.516bn	**£4.530bn**
Animal feed	**£3.504bn**	£1.280bn	**£2.224bn**
Dairy products and eggs	**£4.576bn**	£2.505bn	**£2.071bn**

*Calculated as imports minus exports; values are current prices and rounded. SITC groups include processed goods and re-exports, so the gap is not a directly achievable domestic growing market. Source: Defra / HMRC [1].



Interpretation for Agrispot

- Fruit and vegetables lead the list. This is the strongest category for selective production expansion because several high-import crops can be grown in British fields or protected systems.
- Coffee and cocoa are large but misleading as CEA targets. The statistical group bundles products across raw commodities and processed foods; it identifies exposure, not a viable domestic crop programme.
- The arable and livestock gaps need different levers. More vertical farming is not a credible response to cereal or meat import dependence; field productivity, feed, genetics, processing and trade relationships matter more.

CROP SELECTION / 03

Match the technology to the crop

Agrispot should avoid a one-technology strategy. Greenhouses use natural light and are generally suited to taller, fruiting crops; hydroponic systems are a nutrient and water-delivery method that can be used in either greenhouses or indoor farms; vertical farming adds expensive stacked space and artificial lighting. These are related, but not interchangeable, investment choices.

CROP / CATEGORY	PREFERRED SYSTEM	UK SCALE POTENTIAL	AGRISPOT VIEW
Tomatoes	Glasshouse + substrate hydroponics	**High**	Core target; scale through proven varieties, yields and contracts.
Cucumbers	Glasshouse + substrate hydroponics	**High**	Core target where buyers can take consistent weekly volumes.
Peppers / aubergines	Glasshouse	**Medium-high**	Selective expansion; test energy intensity and wholesale margins.
Strawberries / raspberries	Polytunnels / glasshouse / substrate	**High**	Priority for season extension, resilience and premium quality.
Lettuce / baby leaves	Hydroponic greenhouse; selective vertical	**Medium-high**	Vertical farming only if energy and retail price pencil out.
Basil / coriander / mint	Hydroponic greenhouse or vertical	**Medium-high**	Good candidates for short cycles and freshness-sensitive buyers.
Mushrooms	Controlled growing rooms	**Medium-high**	Distinct model; optimise substrate, hygiene, labour and distribution.
Microgreens / niche herbs	Small vertical farm	**Niche**	Premium contracts first; avoid commodity-scale assumptions.
Coffee / cocoa	Small experimental greenhouse only	**Low for mass-market supply**	Not a core production bet; prioritise sourcing and domestic processing.
Wheat / oats / maize	Conventional field agriculture	**Not a vertical-farm target**	Large biomass, low value per unit weight; improve field systems instead.

A simple rule of thumb

Use the least costly system that delivers the required product specification

For tomatoes, cucumbers and many berries, daylight plus protected cultivation is usually a better starting point than fully enclosed vertical farming. Reserve multi-tier indoor systems for short-cycle, high-value crops where proximity, consistency or quality can offset lighting and climate-control costs.

Not a promise of profitability

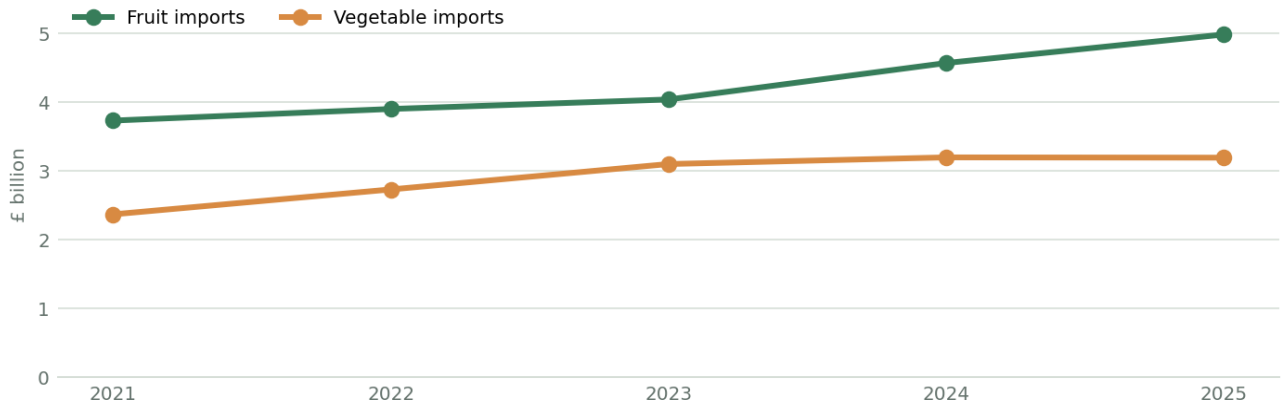
The “scale potential” ratings are strategic screening judgements, not forecasts. Local energy tariffs, site conditions, labour, yields, water security, buyer specifications and finance terms can change a project’s ranking. Each proposed facility requires a crop-specific model and commercial diligence.

CORE INVESTMENT / 04

Protected vegetables and soft fruit

The strongest near-term programme is an expansion of productive British horticulture: glasshouses and tunnels for crops where controlled conditions can improve supply timing, quality and reliability. This builds on existing know-how rather than asking investors to back unproven mass-market tropical crop production.

Fresh fruit and vegetable imports, £bn



Trade in fresh fruit and vegetables, 2025

INDICATOR	2025 RESULT	INVESTMENT IMPLICATION
Vegetable imports	**£3.194bn; 2.113m tonnes**	Opportunity is crop- and season-specific; home production already meets 56% of total vegetable supply.
Fruit imports	**£4.985bn; 3.569m tonnes**	Domestic fruit production met 14% of total supply; prioritise UK-compatible fruit, not tropical/citrus replacement.
Protected vegetables	**258,000 tonnes; £431m value**	An established sector provides a base to modernise and expand, but value fell 1.1% in 2025.
Soft fruit	**134,000 tonnes; £776m value**	High-value segment; 2025 production fell 2.7%, illustrating weather and seasonality exposure.
Strawberries	**93,000 tonnes; £454m value**	Material established category; heat compressed picking windows and caused a later supply shortfall.

Source: Defra, Horticulture Statistics 2025 [2]. Values and volume definitions follow the official release; fruit/vegetable import figures include broad fresh-produce categories and should not be mistaken for the addressable CEA market.

What an investable project looks like

- Buyer-led: target supermarket, wholesaler or food-service demand before committing full capex; negotiate seasonal volume windows rather than assuming a permanent premium.
- Climate- and water-resilient: recirculate irrigation where technically appropriate, secure water rights/supply and provide for storage; protected growing does not remove drought risk.
- Energy-integrated: model heating, supplementary lighting, pumps, cooling and thermal storage together; test base, downside and energy-price shock cases.
- Operationally credible: include crop scheduling, pollination, pest management, labour availability, grading, cold chain and produce waste from the start.

Why the seasonal lens matters

Imports will not disappear simply because domestic output grows. The highest-value opportunity may be a specific seasonal window when imports are costly or supply is unreliable - not year-round substitution at any price.

TECHNOLOGY ECONOMICS / 05

Where hydroponics and vertical farming earn their keep

Hydroponics is a growing method, not a guarantee of lower cost. Vertical farming is an infrastructure-intensive way to control light, temperature, humidity and nutrient delivery within stacked growing layers. The business case depends on what the crop can sell for and how much energy, labour and capital it consumes.

Preferred use cases

USE CASE	WHY IT CAN WORK	MAIN ECONOMIC RISK
Hydroponic greenhouse	Natural daylight; precise irrigation/fertigation; suits fruiting vegetables and herbs.	Heat, glazing and seasonal yield swings; crop disease can affect a concentrated site.
Vertical farm: leafy salads	Fast cycles, repeatable output, proximity to large buyers and reduced weather exposure.	Electricity for lighting and cooling; price pressure from conventional supply.
Vertical farm: herbs / microgreens	Higher value per kg; short delivery routes and consistent specifications can matter.	Demand may be niche; buyer concentration and spoilage can erode margin.
Controlled rooms: mushrooms	Light is not the principal input; production can use suitable substrates and compact rooms.	Substrate quality, hygiene, climate control, labour and batch contamination.

Agrispot investment screen: five tests before capex

1. Unit economics. Model cost per saleable kilogram, not gross harvest. Include reject rates, packaging, labour, nutrients, maintenance, depreciation, finance and distribution.

2. Energy sensitivity. Stress-test electricity and heat prices, winter production, lighting hours and equipment downtime. A facility that only works on optimistic power assumptions is not robust.

3. Offtake confidence. Secure buyer specifications, minimum volumes, price mechanisms and reject standards. Avoid relying on a generic “local” or sustainability premium.

4. Water and site fit. Confirm water quality, supply, discharge requirements, planning, grid connection, logistics and any heat source’s temperature and seasonal availability.

5. Reproducible operations. Prove crop recipes, disease control, automation, harvest labour, cold-chain integration and repeatable saleable yield at pilot scale before copying the design.

Do not confuse tonnes with value

A vertical farm may produce impressive yields per square metre but still be uneconomic per kilogram once electricity and capital charges are included. Agrispot should evaluate profit contribution, cash payback and resilience against a realistic open-field or greenhouse benchmark.

Suggested performance dashboard

CROP	PRODUCTION KPI	FINANCIAL / COMMERCIAL KPI
All CEA crops	Saleable yield; crop cycle; water per kg; reject rate	Gross margin/kg; energy cost/kg; buyer on-time service; waste %
Vertical salad / herbs	kWh/kg; growth days; output per rack / m^2	Contracted volume share; price realisation; payback under downside case
Greenhouse fruiting crops	kg/m^2; seasonal profile; heat input/kg	Seasonal price spread; labour/kg; distribution cost/kg

STRATEGIC IMPORTS / 06

Coffee and cocoa: resilience before cultivation

The combined UK import category “coffee, tea, cocoa, spices, etc.” was valued at £7.552bn in 2025, up 25.7% in current prices from 2024. Exports from the category were £2.071bn. This is a large and volatile category, but it bundles raw agricultural goods with a diverse set of processed and traded products. It is not the value of raw coffee and cocoa beans alone. [1]

Coffee beans

Commercial coffee is produced in tropical and subtropical regions. A British greenhouse would need to provide suitable temperatures, humidity and, during low-light periods, potentially costly supplementary lighting. Coffee trees are perennial, with long lead times and crop cycles very different from fast-growing salad crops. Technical cultivation may be possible, but large-scale production is unlikely to compete with established coffee-growing regions on cost in the near term.

Recommended route for Agrispot

- Support diversified sourcing across producing countries and suppliers; avoid over-dependence on a single climate- or logistics-exposed origin.
- Assess UK value capture in roasting, blending, decaffeination, packaging, product development and premium private-label manufacturing.
- Keep British coffee-growing to small research, demonstration or premium-niche trials until quality, annual yield and full cost/kg are independently verified.

Cocoa beans

Cocoa is a tropical tree crop with long establishment periods and climate-, disease- and pollination-sensitive production. Defra’s 2025 Food Security Digest notes that around two-thirds of global cocoa production is concentrated in West Africa and describes the increased price volatility and climate/plant-health risks affecting the sector. [3] Replicating a productive cocoa agro-ecosystem under British glass would require substantial control of environmental conditions, while competing against regions with established production systems.

Recommended route for Agrispot

- Build supply assurance and risk-management arrangements with more than one origin and appropriately traceable suppliers.
- Explore UK chocolate manufacture and ingredient processing as a domestic value-added opportunity, while recognising that the cocoa bean still has to be imported.
- Research selected alternative ingredients from British oats, pulses and other crops, or fermentation-derived inputs, only where taste, texture, labelling, regulation and consumer acceptance are proven.

> **Investment verdict**
> Do not allocate core greenhouse or vertical-farm capital to mass-market coffee or cocoa production on the basis of import value alone. Treat both as supply-chain resilience and downstream processing opportunities; fund cultivation only as ring-fenced research with explicit stop/go criteria.

Other tropical and high-import crops

The same principle applies to bananas, citrus and many spices: maintain diversified trade relationships where overseas comparative advantage is structural. British production might serve a specialist niche, but heating, lighting, slow plant cycles and scale usually make it an inferior substitute for imports.

EXECUTION / 07

A 24-month Agrispot action plan

Agrispot should build an investable project pipeline rather than launch a blanket subsidy or infrastructure programme. The first objective is to identify projects with verified buyers, robust energy assumptions and crop-specific operating plans.

PHASE	ACTIONS	OUTPUT / GATE
0-3 months Market definition	Map imports at crop and seasonal level; interview buyers; identify import windows, quality specifications and distribution hubs. Screen land, water, grid and heat availability.	Shortlist of 3-5 crops and regions, with addressable market measured in tonnes and seasonal pricing.
3-6 months Pre-feasibility	Develop reference designs for greenhouse, tunnel, hydroponic and vertical options. Obtain indicative capex, energy, labour, yield and packaging assumptions.	Crop-by-crop financial models; downside cases; initial buyer offtake discussions.
6-12 months Pilot and contract	Run pilot crop cycles or work with existing operators; verify yield, quality, waste, crop timing and labour needs. Negotiate buyer contracts and financing.	Evidence-backed go/no-go decision; defined operating procedures and signed or near-final offtake.
12-24 months Scale selectively	Finance and build projects that pass hurdles. Use staged capacity, crop diversification where agronomically sensible, and common data standards across sites.	Replicable commercial facilities; tracked import substitution, unit economics and supply reliability.

Core Agrispot diligence questions

- What exact imported product and season are we replacing? Define variety, grade, packaging and delivery window.
- What is the delivered cost per saleable kilogram? Include capital charges, energy, labour, losses and logistics; compare it with import parity and contract price.
- How much demand is contractable? Obtain volume profiles, rejection rules and pricing formulae from prospective buyers.
- How does the project behave in a downside case? Stress-test energy prices, crop yields, labour costs, interest rates, and 10-20% lower realised prices as scenario assumptions, not forecasts.
- Does the project improve resilience without transferring risk? Check water access, disease exposure, input supply and concentration across customers / sites.

Recommended investment gates

GATE	PASS CONDITION
Market	A clearly defined product and season, addressable volume and credible buyer interest.
Technology	Proven growing protocol with target saleable yield and quality, evidenced by pilot or comparable commercial operations.
Economics	Positive operating contribution and acceptable investment return under realistic downside assumptions; no dependence on a permanent energy-price subsidy.
Resilience	Secure water, energy, key inputs, logistics and crop protection plan.
Scale	Operational playbook, skilled team and repeatable supply chain; expansion staged against demonstrated performance.

Conclusion

Agrispot's best route to commercial impact is a portfolio of targeted growing systems, not an all-purpose "vertical farm" solution. Prioritise protected vegetables and soft fruit; add hydroponic salads and herbs where a verified customer and energy case exists; treat coffee and cocoa principally as import-risk and processing opportunities. Measure success in profitable saleable output and demonstrable import displacement.

APPENDIX / 08

Sources, definitions and limitations

This report uses official UK Government statistics available on 13 October 2026. Data are presented in current prices unless otherwise stated. Recommendations and technology rankings are Agrispot strategic judgements, not official forecasts or guarantees of commercial performance.

Principal sources

[1] Department for Environment, Food & Rural Affairs (Defra). UK trade in Food, Feed and Drink, 2025 updated. Published 8 October 2026. The final 2025 release uses HMRC overseas trade data and ONS trade deflators. It reports £67.8bn imports and £25.7bn exports, and provides current-price SITC category tables.

https://www.gov.uk/government/statistics/uk-trade-in-food-feed-and-drink-2025-updated/uk-trade-in-food-feed-and-drink-2025-updated

[2] Defra. Horticulture Statistics 2025. Updated 25 June 2026. Provides UK horticultural production, supply, value and trade including fresh fruit and vegetables, protected vegetables and soft fruit.

https://www.gov.uk/government/statistics/latest-horticulture-statistics/horticulture-statistics-2025

[3] Defra. United Kingdom Food Security Digest 2025. Published 11 December 2025. Discusses climate, production concentration, supply shocks and price volatility in coffee and cocoa.

https://www.gov.uk/government/statistics/united-kingdom-food-security-digest-2025/united-kingdom-food-security-digest-2025

Definitions and interpretation

- Food, Feed and Drink (FFD): a broad trade category covering multiple agricultural and processed products. The headline £42.1bn net gap is imports less exports in current prices; it is not an addressable CEA market.
- Fruit and vegetables: the Defra trade category includes an extensive range of products. Fresh fruit and vegetable import values in the horticulture release are a separate, more specific series and are not directly interchangeable with the broader FFD category.
- Net import gap: commodity group imports minus exports. It can include processed goods, re-exports and product types not feasible to produce in Britain.
- Protected horticulture: crops grown in glasshouses or polytunnels. Hydroponics refers to a nutrient delivery / growing method; vertical farming generally refers to stacked indoor production with artificial lighting.
- Import substitution: the potential for domestic output to replace a product otherwise imported. Additional output will not automatically reduce imports if demand grows or domestic produce displaces other UK production.

Important limitation

This report is an investment-screening document, not a site-specific feasibility study. It does not estimate project capex, yields, energy consumption, returns or the achievable share of import displacement for any proposed facility. Before investment, each project requires independent agronomic, technical, market, legal, environmental and financial diligence.

AGRISPOT / STRATEGIC RESEARCH

Build where Britain can grow competitively. Import where global comparative advantage remains compelling. Capture more value through technology, contracts and efficient domestic processing.